Identifying Monetary Policy Shocks: A Natural Language Approach
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Contents

Motivation and Main Contribution

Main Contribution

1) Contribute to the literature that identifies monetary policy shocks by applying natural language processing and machine learning
2) Contribute to the literature that applies textual analysis or machine learning to documents produced by the Fed. Creating 'aspect-based' sentiments.

Model definition

The Central Bank set policy instrument s_t:

$$s_t = f(\Omega_t) + \varepsilon_t$$

Ω_t: information set of the Central Bank

$f(\cdot)$: systematic component of monetary policy (endogenous)

ε_t: monetary policy shock

Methodology

Traditional vs Author's approach

1) Traditional Approach:

$$\Delta i_t = \alpha + \beta i_{t-1} + \gamma X_t + {\varepsilon_t}^{RR}$$

i_t: FOMC's FFR target

X_t: Forecasts of the US economy by Fed at time t

$\hat{\varepsilon}_t^{RR}$: Empirical measure for ε_t

2) Authors' Approach:

Natural Language Processing (NLP) and Machine Learning (ML) techniques

Author's approach continued

1) Use NLP to fully capture the systematic component of monetary policy by including valuable information contained in Human language in the economic outlook of Fed economists
2) Include higher-order terms in the econometric model to account for the potential presence of nonlinearities in $f(\cdot)$
3) Use ML techniques to cope with the dimensionality of the problem

Steps of Estimation

Step 1: Process the text of relevant FOMC meeting documents

Step 2: Identify frequently discussed economic concepts in these documents

Step 3: Construct sentiment indicators for each economic concept

Step 4: Run a regression that includes sentiment indicators and numerical forecasts, both linearly and non-linearly

Step 1

Retrieve historical documents (October 1985-2016) associated with FOMC meetings to capture information at the **onset of the meeting**

- Examples of documents included: *Greenbook 1*, *Greenbook 2*, *Tealbook A*, *Redbook*, etc
- Volumn: **772** pdf files for **276** meetings

Read and process the raw textual content for each document

- Remove top words (e.g.:*the*,*is*,*on*,...), numbers (e.g.:dates, page numbers), and "erroneous" words
- Retrieve *singles* (e.g.:inflation), *doubles*, and *triples* (e.g.:consumer price inflation)
- 18,000 singles, 450,000 doubles, and 600,000 triples and calculate the frequency

Step 2

Rank all singles, doubles, and triples by total frequency and identify those that are economic concepts (e.g.:*credit*, *output gap*, etc)

- Use a precise selection algorithm to address the issue of overlap across singles, doubles, and triples that are economic concepts (e.g.:"Commercial real estate" and "real estate")

Figure 1: ECONOMIC CONCEPTS MENTIONED FREQUENTLY IN FOMC DOCUMENTS

Step 3

Capture the sentiment around the **296** individual economic concepts

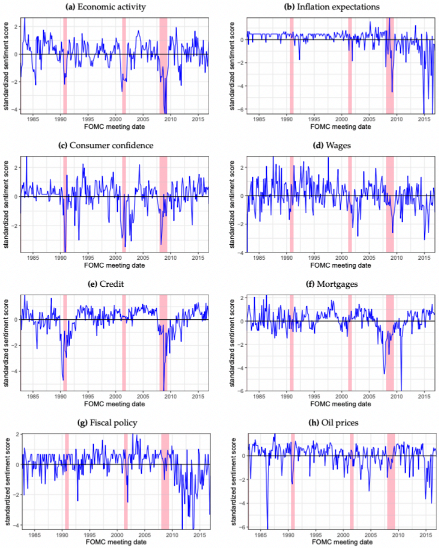
- Check any of the **10** words mentioned before and after the concept's occurrence are associated with positive or negative sentiment (e.g.: able, best, gain, easier for positive sentiments; abandon, bad, frail, negative for negative sentiments)
- Enhance the dictionary of positive and negative terms in Loughran and McDonald (2011)
- Positive word adds a score of $+1$ and each negative word adds a score of -1 towards the sentiment of the concept
- Sentiment indicator for each economic term $=$

$$\frac{\text{Sum of the sentiment scores within the FOMC meeting document}}{\text{Total number of words in the document}}$$

- Obtain time series for each economic concept, where the time variation is across FOMC meetings

Selected Sentiment Indicators

Figure 2: SELECTED SENTIMENT INDICATORS



Step 4

Specify the nonlinear empirical model

$$\Delta i_t = \alpha + \beta i_{t-1} + \Gamma(\widetilde{X}_t, Z_t) + \varepsilon_t *$$

Δi_t: Changes in FOMC's FFR target

$\widetilde{X}_t$: Augmented set of forecasts that Fed economists produce

Z_t: **296** sentiment indicators

$\Gamma(\cdot)$: Non-linear mapping (A second-order polynomial)

Problem: 858 items on the RHS

Step 4 Cont.

Solution: employ a ridge regression to estimate

$$\Delta i_t = \alpha + \beta i_{t-1} + \Gamma(\widetilde{X}_t, Z_t) + \varepsilon_t *$$

$$y_i = \gamma_1 x_{i1} + \cdots + \gamma_k x_{ik} + \varepsilon_i$$

The ridge regression minimizes

$$\sum_i \varepsilon^2{}_i + \lambda \sum_j^k \gamma_j^2$$

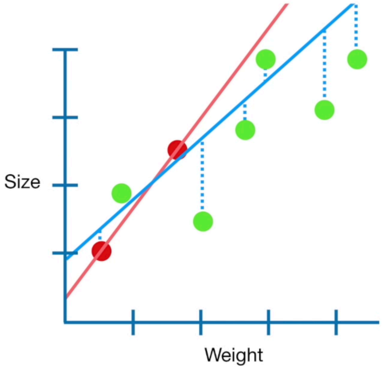
λ: scale of the prior variance

A prior on each coefficient γ that is normally distributed with expected value 0.

Step 4 Cont.

When to use ridge regression?

- a data set contains a higher number of predictor variables than the number of observations
- eliminate multicollinearity in data models



Step 4 Cont.

Two ways to pick λ

- Optimal tuning parameter using k-fold cross-validation
  - Randomly divide sample into k subsamples
  - Use each subsample to fit model on k-1 other subsamples
  - Compute a mean-squared error(MSE) and compute the average MSE
  - Find the smallest MSE by changing λ
- Impose a restriction about the contribution of systematic policy
  - Attribute **90%** of FFR variations to systematic changes and **10%** to shocks

Main Results

Main Results

R^2 for different ML Techniques and Sentiment versions

Table 2: R^2 FOR DIFFERENT MACHINE LEARNING TECHNIQUES AND SENTIMENT VERSIONS

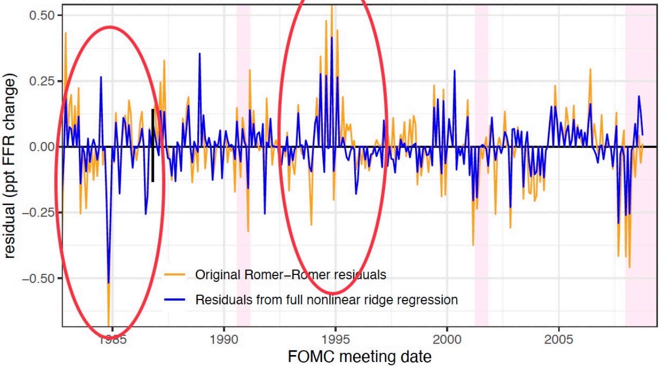
	(1) **10-word sentiment** **Ridge regression**	(2) **5-word sentiment** **Ridge regression**	(3) **10-word sentiment** **LASSO regression**	(4) **5-word sentiment** **LASSO regression**
Romer-Romer OLS	0.50	0.50	0.50	0.50
Romer-Romer ML	0.55	0.55	0.57	0.57
Full linear ML	0.65	0.66	0.55	0.63
Full nonlinear ML	0.75	0.77	0.81	0.72
Full nonlinear ML (90)	0.90	0.90	0.90	0.90

Notes. Implied R^2 from estimating different empirical specifications of equation (3). Column (1) is our preferred model. Column (2) uses a 5-word rather than 10-word distance in Step 3 of our procedure. Columns (3) and (4) employ LASSO rather than ridge regression in Step 4.

Main Results Cont.

What are monetary policy shocks?

Figure 4: ESTIMATED MONETARY POLICY SHOCKS



Notes. Time series of estimated monetary policy shocks. Dark blue: our preferred version, the residuals from predicting changes in the FFR based on numerical forecasts, sentiment indicators and nonlinearities in FOMC documents. Orange: benchmark version based on a specification that follows Romer and Romer (2004). Shaded areas represent NBER recessions.

Conclusion

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- Develop a method to identify monetary policy shocks using NLP and ML
- Extract and develop sentiment indicators for 296 economic concepts
- Include the Fed economists' numerical forecasts of macroeconomic variables, as well as the sentiment indicators in a ridge regression to predict systematic changes in the target interest rate
- Deliver a cleanly estimated series of monetary policy shocks

Thank you!